

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and owning end-to-end distributed systems at scale. Built and operated **YouTube** real-time analytics infrastructure processing billions of events daily and led **Google** search ranking systems serving billions of queries. Architected **C++** low-latency market systems for **Whatnot** and managed live streaming infrastructure handling 10x traffic spikes. Collaborated across engineering, product, and research teams to deliver measurable improvements in reliability and performance. Works on data pipeline architecture, distributed systems design, and on-call excellence. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: C++, Python, Go, SQL, TypeScript, JavaScript, HTML, CSS, Java

Distributed Systems: Distributed Systems, Microservices, Real-time Data Processing, Scalability, Load Balancing

Data Infrastructure: Data Pipelines, Kafka, Data Processing, Big Data, Analytics

Cloud & Infrastructure: GCP, Kubernetes, Linux, On-call Operations

Leadership: Technical Leadership, System Architecture, Mentorship, Cross-team Collaboration

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Built and maintained data pipelines ingesting user interaction signals to support A/B experimentation platform serving 2 billion monthly active users.
- Designed real-time analytics infrastructure processing billions of daily events with sub-second latency for creator dashboard metrics and recommendation features.
- Partnered with product and research teams to define and deliver data quality metrics for content understanding systems.
- Established on-call rotation and incident response protocols for analytics services, reducing mean time to resolution by 40%.
- Conducted technical interviews and mentored senior engineers on distributed systems design and data infrastructure best practices.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Designed and deployed real-time inventory and pricing data pipelines using **Kafka** and stream processing to support marketplace search and recommendations.
- Architected low-latency market systems in **C++** handling live bid updates for 1M+ concurrent users during peak auction events.
- Collaborated with product and data science teams to define real-time metrics for seller analytics and buyer personalization features.
- Optimized **Scalability** of order matching infrastructure, supporting 10x traffic spikes during high-demand drops without degradation.
- Led architecture reviews for microservices migration, improving deployment frequency and system reliability across the platform.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Led a team of 8 engineers owning **YouTube** search ranking infrastructure, delivering measurable improvements in watch time and engagement metrics.
- Designed **Distributed Systems** for content indexing and retrieval, processing billions of videos with query latency under 100ms.

- Owned end-to-end reliability for core search services, achieving 99.99% uptime and reducing on-call incident volume by 60%.
- Partnered with **Machine Learning** research teams to deploy ranking models serving billions of daily queries in production.
- Mentored engineers through promotion cycles and conducted technical deep-dives on **System Architecture** and performance optimization.

Software Engineer | Google | 02/2011 - 02/2015

- Designed and implemented monitoring and alerting systems for search infrastructure, reducing detection time for traffic anomalies.
- Collaborated with search quality team to integrate new ranking signals into serving infrastructure at massive scale.
- Built **Data Processing** workflows for query log analysis, supporting product decisions for search ranking experiments.
- Developed core infrastructure components for **Google Search** indexing pipeline, improving crawl efficiency and freshness.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Designed and shipped features for **Windows Live Messenger** infrastructure serving millions of concurrent users in **C++**.
- Conducted performance profiling and optimization of core messaging pipeline, reducing server resource utilization by 25%.
- Collaborated with program management to define technical requirements and delivery timelines for release milestones.
- Implemented streaming components for **Silverlight** media delivery platform, improving playback initialization time.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed and tested features for **Windows Live Messenger** client, contributing to shipped product functionality.
- Participated in code reviews and learned **C++** development practices within a large-scale engineering organization.

Education

B.S. Honors Degrees in Computer Science

Virginia Tech | 2004 - 2007